Activity 12

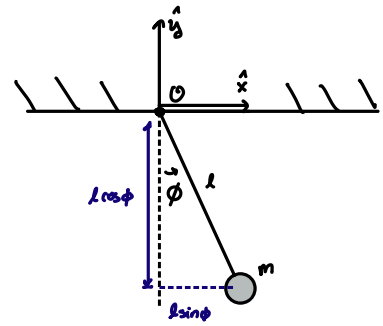


Figure 1: A pendulum: a rope of fixed length l is attached to the ceiling and a mass m.

1. Consider the pendulum shown in Fig. 1. The pendulum only moves in the x-y plane. Write down the Lagrangian for this pendulum in terms of the minimum number of generalized coordinates and find the equation of motion for the pendulum.

Position of pendulum:

$\vec{r} = l\sin\phi\,\hat{x} - l\cos\phi\,\hat{y}$

$\vec{v} = l\cos\phi\,\dot{\phi}\,\hat{x} + l\sin\phi\,\dot{\phi}\,\hat{y}$

$T = \frac{1}{2}m\left(l^2\cos^2\phi\,\dot{\phi}^2 + l^2\sin^2\phi\,\dot{\phi}^2\right) = \frac{1}{2}ml^2\dot{\phi}^2$

$U = mg(-l\cos\phi) = -mgl\cos\phi$

$\Rightarrow \mathcal{L} = \frac{1}{2}ml^2\dot{\phi}^2 + mgl\cos\phi$

$\frac{\partial \mathcal{L}}{\partial \phi} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\phi}}$

$\Rightarrow \frac{\partial \mathcal{L}}{\partial \phi} = -mgl\sin\phi$

$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = ml^2\dot{\phi}$, $\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = ml^2\ddot{\phi}$

$\Rightarrow -mgl\sin\phi = ml^2\ddot{\phi}$

$\boxed{g\sin\phi = l\ddot{\phi}}$

2. Now let's approach the pendulum problem using Lagrange multipliers. First, use the constraint condition $f(x, y) = \sqrt{x^2 + y^2} = l = const.$

   (a) Write the Lagrangian in (x, y) coordinates.

   $\mathcal{L} = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{y}^2 - mgy$

   (b) The modified Lagrangian equations including a Lagrange multiplier λ are:

$$\frac{\partial \mathcal{L}}{\partial x_i} + \lambda\frac{\partial f}{\partial x_i} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}_i}, \qquad f(x_1, ..., x_N) = const. \tag{1}$$

   Use this to find a set of differential equations that include $\lambda(t)$.

$\frac{\partial \mathcal{L}}{\partial x} + \lambda\frac{\partial f}{\partial x} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}}$

$0 + \lambda\frac{1}{2}(x^2+y^2)^{-1/2}(2x) = \frac{d}{dt}(m\dot{x}) = m\ddot{x}$

$l = \sqrt{x^2+y^2}$

$\frac{\lambda x}{\sqrt{x^2+y^2}} = m\ddot{x} \Rightarrow \boxed{\frac{\lambda x}{l} = m\ddot{x}}$

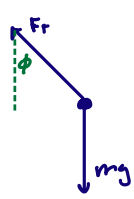
$\frac{\partial \mathcal{L}}{\partial y} + \lambda\frac{\partial f(x,y)}{\partial y} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}}$

$-mg + \lambda\frac{1}{2}(x^2+y^2)^{-1/2}\,2y = m\ddot{y}$

$-mg + \frac{\lambda y}{\sqrt{x^2+y^2}} = m\ddot{y} = \boxed{-mg + \frac{\lambda y}{l} = m\ddot{y}}$

   (c) Draw a free-body diagram for the pendulum, and using Newton's second law, verify that:

$$\lambda\frac{\partial f}{\partial x} = F_x^{\text{cstr}} \qquad \lambda\frac{\partial f}{\partial y} = F_y^{\text{cstr}}$$



$F_{Ty} = m\ddot{x} = \frac{\lambda x}{l} = \lambda\frac{\partial f}{\partial x}$ ☑

$-mg + F_{Ty} = m\ddot{y}$

$F_{Ty} = m\ddot{y} + mg = \frac{\lambda y}{l} = \lambda\frac{\partial f}{\partial y}$ ☑

   (d) Find λ in terms of F_T, the tension in the rope

$F_T = \sqrt{F_{Tx}^2 + F_{Ty}^2} = \sqrt{\left(\frac{\lambda x}{l}\right)^2 + \left(\frac{\lambda y}{l}\right)^2} = \sqrt{\frac{\lambda^2}{l^2}}\sqrt{x^2+y^2} = \frac{\lambda}{l}\sqrt{x^2+y^2} = \frac{\lambda}{l}(l) = \boxed{\lambda}$

3. Consider the same system, but now use the constraint condition $f(x, y) = x^2 + y^2 = l^2 = const.$

   (a) Write down the new modified Lagrange equations including the Lagrange multiplier

$$\frac{\partial L}{\partial x} + \lambda \frac{\partial f}{\partial x} = \frac{d}{dt}\frac{\partial L}{\partial \dot{x}}$$

$$\frac{\partial L}{\partial x} = 0 \qquad \frac{\partial f}{\partial x} = 2x$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = \frac{d}{dt}(m\dot{x}) = m\ddot{x}$$

$$\Rightarrow \boxed{2\lambda x = m\ddot{x}}$$

$$\frac{\partial L}{\partial y} + \lambda \frac{\partial f}{\partial y} = \frac{d}{dt}\frac{\partial L}{\partial \dot{y}}$$

$$-mg + \lambda(2y) = \frac{d}{dt}(m\dot{y})$$

$$\boxed{-mg + 2\lambda y = m\ddot{y}}$$

   (b) Using your free body diagram and Newton's second law, verify that

$$\lambda \frac{\partial f}{\partial x} = F_x^{\text{cstr}} \qquad\qquad \lambda \frac{\partial f}{\partial y} = F_y^{\text{cstr}}$$

$$\frac{\lambda \partial f}{\partial x} = 2\lambda x = m\ddot{x} \overset{\text{from } 2(c)}{=} F_{Tx} \quad ☑$$

$$\frac{\lambda \partial f}{\partial y} = 2\lambda y = m\ddot{y} + mg \overset{\text{from } 2(c)}{=} (F_{Ty} - mg) + mg = F_{Ty} \quad ☑$$

   (c) Find λ in terms of F_T, the tension in the rope

$$F_T = \sqrt{F_{Tx}^2 + F_{Ty}^2} = \sqrt{(2\lambda x)^2 + (2\lambda y)^2} = 2\lambda\sqrt{x^2 + y^2} = 2\lambda l$$